Measuring Inequality without the Gini

A Displacement- and Structure-Based Reformulation of the Inequality Channel

Joint Digital Transformation and the Middle-Income Trap — Equations Addendum

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Motivation. The Gini channel of the baseline framework (Eqs. 4′, 9′, 13′, 14′) cannot be identified under country fixed effects: in the assembled panel the disposable Gini has a within-country / between-country standard-deviation ratio of 0.07, so country effects absorb essentially all of its variation and what remains is interpolation noise. This addendum keeps the *economics* of the inequality channel but replaces the *measure* with two quantities the data can actually identify — structural inequality (the dual economy) and displacement inequality (workers pushed out of contracting sectors) — and provides the one non-circular test of whether AI *creates* inequality, on the labor share of income.

1 Structural inequality: the dual economy

Let H_{it} denote human capital and σ_{it}^{KIS} the share of knowledge-intensive services. Define the structural-inequality (dual-economy) index

$$D_{it}^{\text{struct}} \;=\; \sigma_{it}^{KIS}\,\big(1 - H_{it}\big). \tag{S1}$$

D^{struct} is large when a sizeable modern services sector coexists with a low-skill workforce — a productive sector most workers cannot access. Unlike the Gini it varies across countries (between-country s.d. = 0.077) and moderately within (within/between = 0.12). It is the inequality the structural mechanism is really about, and it is the object that defines the low-H/high-services (LH) quadrant where the joint-digital complementarity is concentrated.

The joint-digital growth effect, conditioned on structural inequality:

$$g_{it} = \alpha + \beta_1 F_{it-1} + \beta_2 G_{it-1} + \beta_3 (FG)_{it-1} + \beta_4 (FG \times D^{\text{struct}})_{it-1} + \Gamma' X_{it-1} + \mu_i + \lambda_t + \varepsilon_{it}, \tag{S2}$$

with predicted $\beta_4 > 0$: the FinTech–AI complementarity pays off more in more dual (structurally unequal) economies.

2 Displacement inequality: workers pushed out

The baseline displacement law (Eq. 7) is $D_{it} = \delta_0 G_t \rho_{it}(1 - H_{it})(1 - \sigma_{it}^{KIS})$. Operationalize it with the *time-varying* AI series (publications, normalized within year to $\tilde{G}_{it} \in [0, 1]$) rather than the broadcast AI snapshot, giving the displacement-exposure index

$$D_{it}^{\text{exp}} \;=\; \tilde{G}_{it}\,\big(1 - H_{it}\big)\,\big(1 - \sigma_{it}^{KIS}\big). \tag{D1}$$

Because $\tilde{G}$ moves within country, so does $D^{\exp}$: its within/between ratio is 0.166 (identifiable under FE), versus 0.05 for the snapshot-based displacement term and 0.07 for the Gini. This is the inequality measure that "moves."

Growth with a displacement drag (the η_9 channel of Eq. 14′, with $D^{\exp}$ in place of the Gini-amplified term):

$$g_{it} = \alpha + \beta_1 F_{it-1} + \beta_2 G_{it-1} + \beta_3 (FG)_{it-1} - \beta_5 D^{\exp}_{it-1} + \Gamma' X_{it-1} + \mu_i + \lambda_t + \varepsilon_{it}, \qquad \text{(D2)}$$

predicted $\beta_5 > 0$ (displacement exposure lowers growth), and analogously a positive coefficient in the slowdown LPM.

Identification caveat (important). $D^{\exp}$ is constructed from AI; within country it correlates with G at ≈ 0.88. It is therefore an *exposure / vulnerability* index, not an AI-independent outcome, and in (D2) it cannot be cleanly separated from the direct AI term. Report it as a combined exposure channel, not as a separately identified displacement effect.

3 Does AI *create* inequality? The non-circular test

To ask whether AI generates inequality one needs a distributional *outcome* that (i) moves over time and (ii) is not built from AI. The Gini fails (i); $D^{\exp}$ fails (ii). The labor share of income LS_{it} (PWT `labsh`, = 1− capital share) satisfies both. The first stage of the indirect channel (Eq. 49 / Spec 15), re-expressed on the labor share:

$$LS_{it} = \gamma_0 + \gamma_1 G_{it-1} + \gamma_2 \big[G(1-H)\big]_{it-1} + \gamma_3 F_{it-1} + \Gamma' X_{it-1} + \mu_i + \lambda_t + v_{it}. \qquad \text{(L1)}$$

Hypotheses: $\gamma_1 < 0$ (AI lowers labor's income share ⇒ raises inequality) and $\gamma_2 < 0$ (the effect is larger where human capital is scarce — skill-biased technical change). Because LS moves and is exogenous to the construction of G, (L1) is the clean, non-circular counterpart of the displacement story, and the mediated growth decomposition

$$\frac{\partial g}{\partial G} = \underbrace{\beta_2 + \beta_3 F}_{\text{direct + joint-digital}} + \underbrace{\gamma_1\, \theta}_{\text{indirect via labor share}} \qquad \text{(L2)}$$

identifies the indirect AI-through-inequality drag as $\gamma_1\theta$, with θ the labor-share coefficient in the growth equation $g_{it} = \cdots + \theta\, LS_{it-1} + \ldots$.

4 Empirical results on the assembled panel

All estimates are two-way (country and year) fixed effects with country-clustered standard errors, lagged regressors, on `final_panel_imputed__5_.csv`.

Bottom line. Inequality is visible in this panel through *structure* (S1, the dual-economy map) and through *displacement exposure* (D1, which moves), and both are theory-consistent. But the claim that AI *creates* income inequality is not supported here — it is a precise null within country and confounded between — and the honest way to test it is the labor-share equation (L1), which requires adding PWT `labsh`. Until then, inequality enters the paper as a conditioning structure and an exposure channel, not as an AI-generated outcome.

Quantity	Measure	Coefficient	p	Reading
Within/between variation	Gini (disp.)	0.071	—	frozen
Within/between variation	snapshot D_{term}	0.050	—	frozen
Within/between variation	D^{exp} (this paper)	**0.166**	—	moves
Joint digital × structure	FG (LH quadrant)	+5.80	< 0.001	robust, conditional
Displacement drag	D_l^{exp} on growth	−6.15	0.402	right sign, n.s. (collinear)
Displacement → slowdown	D_l^{exp}	+0.64	0.546	right sign, n.s.
AI → Gini (within)	G_l	−0.008	0.624	null
AI → Gini, skill-bias	$[G(1-H)]_l$	−0.001	0.954	null
AI → Gini (between)	G	−0.028	0.002	wrong sign (development)
AI → labor share	`labsh`	*pending data*		non-circular test (L1)

Table 1: Inequality on the assembled panel. The displacement index is the only inequality measure with usable within-country variation, but it is collinear with AI and so cannot be separated from the direct AI effect; the labor-share test (L1) is the clean route and awaits one PWT column. The Gini cannot show AI creating inequality: within-country the effect is a precise null, and the between-country sign is negative (high-AI economies are richer and less unequal — a development confound, not an AI effect).